

## NITRA, Slovakia

WMO: 118550

Lat: 48.283N

Lon: 18.133E

Elev: 446

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 11.3       | 16.3 | 4.8                           | 7.2 | 14.5 | 9.9 | 9.3 | 20.1 | 31.0                  | 41.9 | 27.9 | 37.4 | 4.1                      | 320  | 0.558 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.3                   | 91.2            | 69.8 | 87.8 | 68.5 | 84.7 | 67.2 | 71.6                | 86.2 | 70.1 | 84.3 | 68.8 | 81.9 | 8.6                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.7                            | 99.8 | 77.5 | 65.3 | 95.2 | 75.9 | 64.1 | 91.0 | 74.4 | 35.6          | 86.2 | 34.3 | 84.4 | 33.2 | 82.4 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |  |  |
| 27.1              | 23.7 | 21.1 | 5.2                        | 95.1 | 7.2                | 3.1 | 0.0                                                | 97.4 | -4.2       | 99.2 | -8.3       | 100.9 | -13.6      | 103.1 |  |  |
|                   |      |      | 4.6                        | 73.9 | 7.1                | 1.5 | -0.5                                               | 74.9 | -4.7       | 75.7 | -8.6       | 76.6  | -13.8      | 77.6  |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.3        | 30.6  | 34.6  | 42.2  | 52.5  | 60.3  | 66.6  | 70.3  | 69.7  | 61.0  | 51.6  | 42.5  | 33.0  | (6)  |
|                                                                        | DBStd       | 15.55       | 7.93  | 7.71  | 7.14  | 7.34  | 6.34  | 6.22  | 5.99  | 6.02  | 6.29  | 7.26  | 7.40  | 7.81  | (7)  |
|                                                                        | HDD50       | 2191        | 603   | 431   | 255   | 57    | 4     | 0     | 0     | 0     | 1     | 70    | 242   | 528   | (8)  |
|                                                                        | HDD65       | 5506        | 1068  | 850   | 706   | 378   | 172   | 54    | 20    | 23    | 150   | 418   | 675   | 992   | (9)  |
|                                                                        | CDD50       | 2674        | 0     | 1     | 14    | 132   | 323   | 497   | 629   | 610   | 333   | 118   | 17    | 0     | (10) |
|                                                                        | CDD65       | 515         | 0     | 0     | 0     | 3     | 26    | 101   | 184   | 168   | 32    | 1     | 0     | 0     | (11) |
|                                                                        | CDH74       | 5726        | 0     | 0     | 0     | 54    | 346   | 1140  | 2073  | 1787  | 316   | 10    | 0     | 0     | (12) |
|                                                                        | CDH80       | 2019        | 0     | 0     | 0     | 5     | 57    | 369   | 834   | 692   | 62    | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 8.7         | 8.9   | 9.4   | 10.1  | 9.3   | 8.8   | 7.9   | 8.0   | 7.3   | 7.9   | 8.2   | 9.7   | 9.0   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 24.2        | 1.5   | 1.4   | 1.4   | 1.7   | 2.6   | 2.6   | 2.8   | 2.2   | 2.2   | 2.0   | 2.0   | 1.8   | (15) |
|                                                                        | PrecMax     | 37.1        | 2.6   | 3.8   | 3.8   | 4.1   | 6.8   | 5.6   | 6.1   | 4.4   | 6.4   | 5.4   | 4.5   | 5.1   | (16) |
|                                                                        | PrecMin     | 15.1        | 0.1   | 0.0   | 0.1   | 0.0   | 0.8   | 0.6   | 0.1   | 0.1   | 0.4   | 0.2   | 0.2   | 0.4   | (17) |
|                                                                        | PrecStd     | 4.3         | 0.7   | 0.9   | 1.0   | 1.0   | 1.4   | 1.2   | 1.5   | 1.3   | 1.4   | 1.2   | 1.0   | 1.1   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 52.0  | 57.3  | 67.9  | 79.7  | 84.9  | 91.7  | 95.0  | 95.9  | 85.8  | 75.1  | 62.8  | 51.8  | (19) |
|                                                                        |             | MCWB        | 46.8  | 48.1  | 52.6  | 60.3  | 65.7  | 70.7  | 70.8  | 70.0  | 66.2  | 61.8  | 55.8  | 47.6  | (20) |
|                                                                        | 2%          | DB          | 46.8  | 52.4  | 62.9  | 74.7  | 81.1  | 87.9  | 91.4  | 91.2  | 81.1  | 70.7  | 58.7  | 48.1  | (21) |
|                                                                        |             | MCWB        | 43.5  | 46.1  | 50.6  | 57.8  | 64.3  | 69.6  | 70.0  | 69.4  | 64.5  | 59.8  | 53.2  | 45.4  | (22) |
|                                                                        | 5%          | DB          | 43.7  | 48.6  | 58.5  | 70.8  | 78.1  | 84.4  | 88.0  | 87.3  | 77.6  | 67.2  | 55.7  | 45.4  | (23) |
|                                                                        |             | MCWB        | 41.1  | 44.1  | 48.5  | 55.9  | 62.9  | 67.9  | 68.2  | 68.4  | 63.0  | 58.5  | 51.0  | 43.1  | (24) |
|                                                                        | 10%         | DB          | 40.9  | 45.7  | 54.5  | 66.8  | 74.4  | 80.8  | 84.7  | 83.6  | 73.7  | 63.3  | 53.1  | 42.8  | (25) |
|                                                                        |             | MCWB        | 38.8  | 42.1  | 46.5  | 54.2  | 61.0  | 66.0  | 67.4  | 66.7  | 61.3  | 56.4  | 49.4  | 40.8  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 47.6  | 49.7  | 54.4  | 61.6  | 68.7  | 74.1  | 72.9  | 73.1  | 68.4  | 63.6  | 56.5  | 48.3  | (27) |
|                                                                        |             | MCDB        | 51.4  | 55.8  | 63.9  | 76.3  | 80.6  | 87.5  | 89.2  | 87.9  | 81.0  | 71.7  | 61.3  | 51.1  | (28) |
|                                                                        | 2%          | WB          | 44.1  | 46.8  | 51.8  | 59.0  | 66.0  | 71.2  | 71.3  | 71.1  | 66.4  | 61.2  | 53.9  | 45.8  | (29) |
|                                                                        |             | MCDB        | 46.3  | 51.3  | 60.5  | 71.9  | 78.2  | 84.2  | 86.7  | 85.9  | 77.3  | 68.7  | 58.1  | 47.7  | (30) |
|                                                                        | 5%          | WB          | 41.2  | 44.4  | 49.6  | 56.9  | 64.1  | 69.4  | 69.9  | 69.7  | 64.6  | 59.2  | 51.8  | 43.2  | (31) |
|                                                                        |             | MCDB        | 43.1  | 48.1  | 57.1  | 68.4  | 75.1  | 81.7  | 84.5  | 84.3  | 73.9  | 65.7  | 55.0  | 45.0  | (32) |
|                                                                        | 10%         | WB          | 38.8  | 42.3  | 47.4  | 55.1  | 62.2  | 67.5  | 68.6  | 68.0  | 62.7  | 57.2  | 49.7  | 40.9  | (33) |
|                                                                        |             | MCDB        | 40.6  | 45.4  | 53.5  | 65.0  | 72.2  | 78.4  | 82.0  | 80.9  | 71.7  | 62.6  | 52.6  | 42.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.9   | 12.7  | 17.0  | 20.4  | 20.7  | 21.4  | 22.3  | 21.9  | 19.3  | 16.5  | 10.8  | 8.6   | (35) |
|                                                                        |             | MCDBR       | 12.1  | 17.5  | 24.6  | 26.8  | 26.9  | 26.9  | 27.9  | 27.5  | 25.6  | 21.7  | 14.6  | 10.3  | (36) |
|                                                                        | 5% WB       | MCWBR       | 9.3   | 12.3  | 14.5  | 13.2  | 12.7  | 11.6  | 10.3  | 10.3  | 11.8  | 12.4  | 9.8   | 8.4   | (37) |
|                                                                        |             | MCWBR       | 11.4  | 15.8  | 21.5  | 24.0  | 23.9  | 23.9  | 25.3  | 24.4  | 21.4  | 19.0  | 12.8  | 9.5   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.324 | 0.347 | 0.384 | 0.431 | 0.427 | 0.427 | 0.438 | 0.437 | 0.410 | 0.381 | 0.358 | 0.323 | (40) |
|                                                                        |             | taud        | 2.346 | 2.319 | 2.251 | 2.147 | 2.204 | 2.242 | 2.211 | 2.228 | 2.271 | 2.327 | 2.343 | 2.372 | (41) |
|                                                                        | Edn at Noon | Edn at Noon | 232   | 253   | 261   | 259   | 266   | 266   | 260   | 255   | 249   | 236   | 216   | 217   | (42) |
|                                                                        |             | Edh at Noon | 23    | 29    | 36    | 44    | 44    | 43    | 43    | 41    | 36    | 29    | 23    | 20    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 308   | 561   | 941   | 1399  | 1666  | 1815  | 1749  | 1568  | 1118  | 682   | 350   | 240   | (44) |
|                                                                        | RadStd      | RadStd      | 43    | 84    | 143   | 142   | 187   | 147   | 161   | 153   | 146   | 80    | 40    | 25    | (45) |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            |       | +1.28   | N/A    | N/A     | +2.30 | N/A   | N/A         | -227  | N/A   | +200  |
| (47) Regional (76 neighbors) |       | +1.06   | N/A    | N/A     | +1.53 | +0.88 | +0.77       | -230  | -319  | +160  |

Nomenclature: See separate page